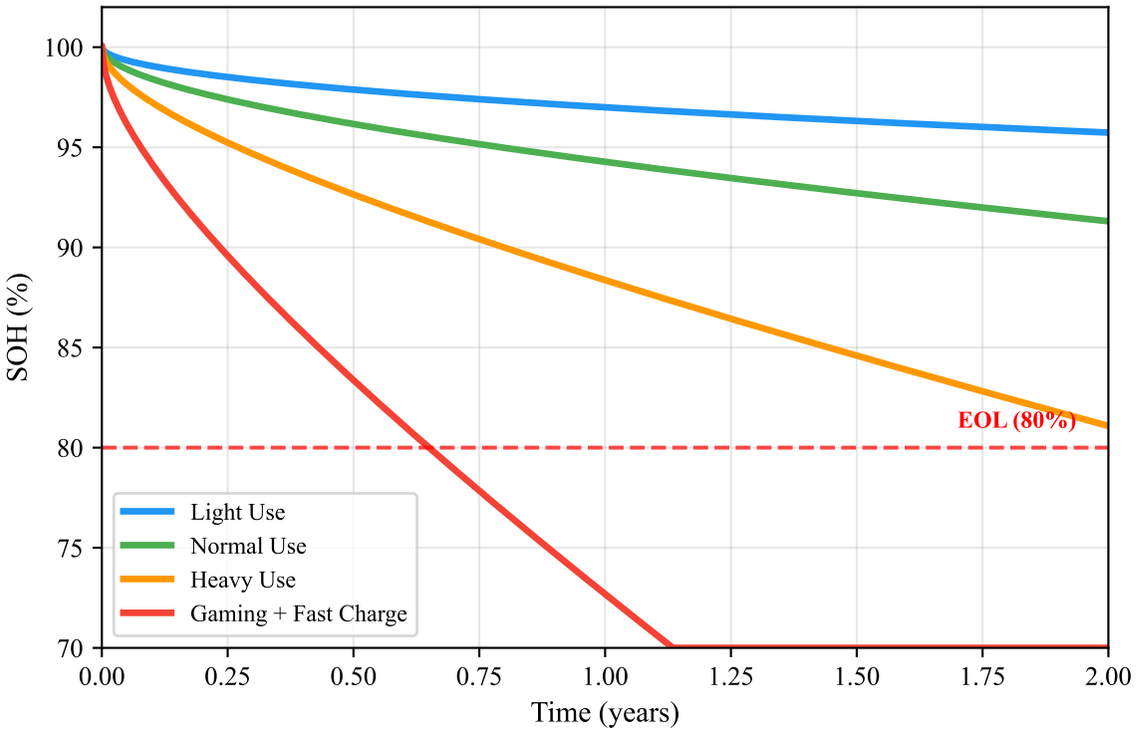
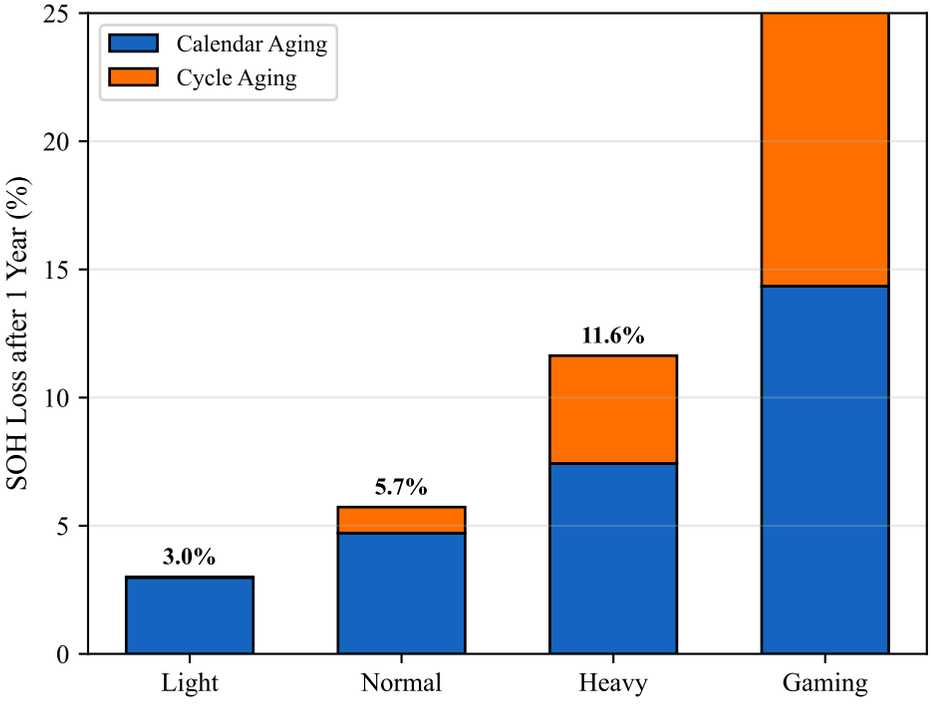
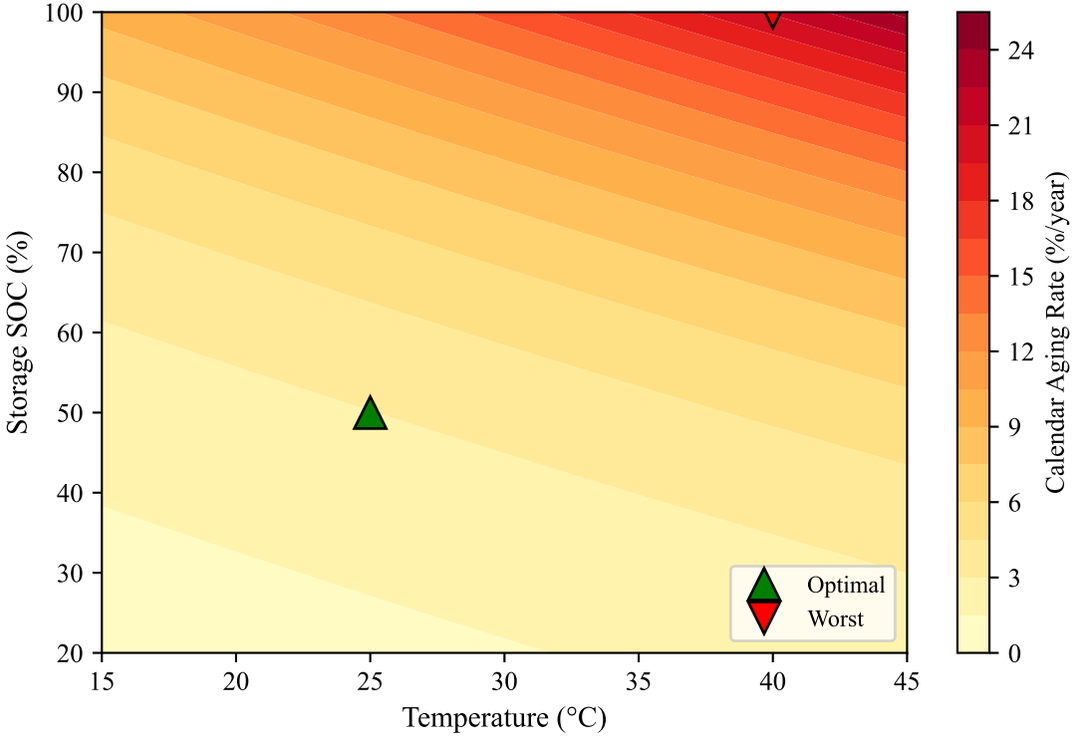






Combined Aging Model

$$\frac{dSOH}{dt} = -\dot{SOH}_{cal} - \dot{SOH}_{cyc}$$

$$\dot{SOH}_{cal} = A_{cal} \cdot e^{-E_a/RT} \cdot e^{\beta(SOC - 0.5)}$$

$$\dot{SOH}_{cyc} = k_{cyc} \cdot DOD^{\gamma} \cdot (I/I_{nom})^{\alpha} \cdot |I|$$

Recommendations :

1. Keep SOC between 20-80%
2. Avoid high temperature (>35C)
3. Use slow charging when possible
4. Avoid deep discharge (<10%)
5. Don't leave at 100% for long

Worst Case :

Gaming while fast charging at high SOC in hot environment